

Ralph & Clawd: Hardware-Isolated AI Gateway

Security Architecture & Sandboxing Report

Executive Summary

This document outlines the security measures implemented to transform Ralph from a local daemon into a multi-user AI gateway. The core philosophy is 'Zero Trust Virtualization', ensuring that even under a total software compromise, the host system remains physically unreachable.

Layer 1: Hardware-Backed Virtualization (OrbStack/Docker)

Unlike software containers that share the host kernel, Ralph is isolated using hardware virtualization (Intel VT-x / Apple Silicon Virtualization). The CPU enforces a hard memory boundary between your Mac and the AI processes.

Layer 2: File System Restricted Mounts

Path	Access	Security Status
/Users/isakzvegelj/clawd	Read/Write	Isolated Project Folder
/Users/isakzvegelj/Documents	NONE	Hardware-Blocked
/Users/isakzvegelj/.ssh	NONE	Hardware-Blocked
/Users/isakzvegelj/Desktop	NONE	Hardware-Blocked

Layer 3: Multi-User Subscription Proxy

Ralph acts as a 'Gatekeeper' for your AI subscriptions. Users connect via Discord, but their requests are queued and processed as distinct 'User Contexts'. Each user's history is isolated in separate memory files, preventing data leaks between friends or clients.

Conclusion

By leveraging hardware virtualization, Ralph is now safe to be used as a public-facing AI server. Your private computer data is physically invisible to the AI, allowing you to safely share your computational power and AI intelligence with others.